American Computer Science League

2021 Finals ● Short Problems ● Senior Division

1. Boolean Algebra

Simplify the following expression:

$$\overline{A\overline{B}C(A + B)(\overline{B} + \overline{C})}$$

A. $A\overline{B}C$
B. $A + \overline{B} + C$
C. $\overline{A}B\overline{C}$
D. $\overline{A} + B + \overline{C}$
E. None of the above

2. Boolean Algebra

How many ordered triples make the following expression TRUE?

$$(\overline{AB + C}) + (\overline{A} * \overline{B} + \overline{C})$$

A. 1
B. 3
C. 5
D. 7
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

((LCIRC-25 011001) OR (RCIRC-16 101101) AND (NOT 011100))

A. 110010
B. 100010
C. 100011
D. 110110
E. None of the above

4. Bit-String Flicking

Which bit string values of X (a bitstring of 5 bits) make the following equation TRUE?

((LSHIFT-1 X) OR (NOT (RSHIFT-1 10111))) = X

A. 11111
B. 11100
C. 1110*
D. 111*0
E. None of the above

5. Recursive Functions

Find $f(20)$ given:

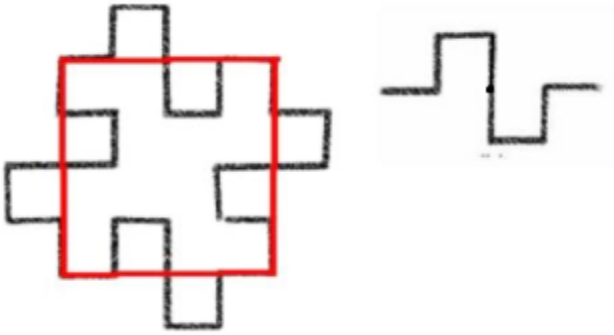
$$f(x) = \begin{cases} f(f(x-3)) + 3 & \text{if } x \geq 10 \\ f(x-2) - 2 & \text{if } 4 < x < 10 \\ x - 1 & \text{if } x \leq 4 \end{cases}$$

A. 1
B. 3
C. 5
D. 7
E. None of the above

6. Recursive Functions

The squareflake curve is generated by beginning with a square (Stage 1), then replacing each side of the square with the figure on the right using 8 smaller segments of the same length to form a new curve (Stage 2).

Continue the process on each segment of the new curve that is created.



The rule for generating the squareflake fractal curve is to replace each line segment with the zig-zag curve consisting of 8 segments shown above on the right.

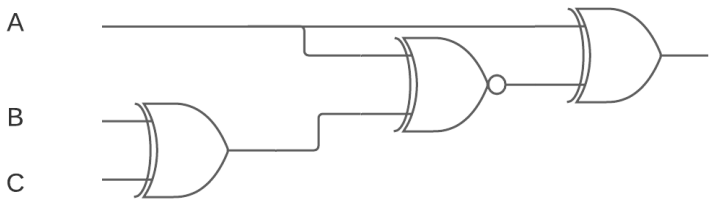
The diagram is taken from https://nrich.maths.org/

If the original red square has a perimeter of length 64, during the construction of this fractal, at which stage will the length of the entire perimeter of the new curve exceed 4000?

A. 5
B. 6
C. 7
D. 8
E. None of the above

7. Digital Electronics

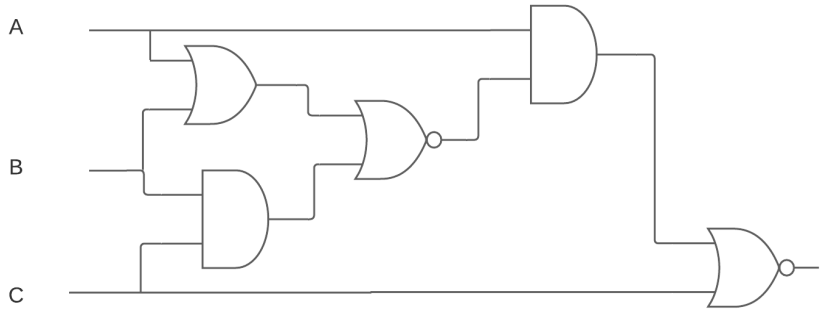
How many ordered triples make the following circuit TRUE?



A. 0
B. 2
C. 4
D. 8
E. None of the above

8. Digital Electronics

Simplify the Boolean expression represented by the digital circuit below:



A. $\overline{A}\ C$
B. $\overline{C}$
C. $B\ C$
D. C
E. None of the above

9. Prefix-Infix-Postfix

Convert the following postfix expression to prefix if & is a trinary operator and ! is a unary operator (numbers are single digits):

3 4 5 + 7 2 1 & 6 ! * * 7 8 - 4 3 + 2 & 3 2 ^ ! / &

A. &3**+45&721!6/& -78 +4 32!^32
B. &3+*4 5*&721!6/&-78 +432!^32
C. &3*+45&*721!6/&-78 +432!^32
D. &3*+45*&721!6/&-78 +432!^32
E. None of the above

10. Prefix-Infix-Postfix

Evaluate this prefix expression if all numbers are single digits:

* - + 4 7 - 1 8 / + ^ 5 2 * 5 - 6 2 ^ 3 2

A. -90
B. -20
C. 90
D. 120
E. None of the above

11. Computer Number Systems

Evaluate the following expression in hexadecimal:

$A74F2C_{16} / 4_{16}$

A. $A74F2B_{16}$
B. $29D3CB_{16}$
C. 12351714_{16}
D. $57A8E2_{16}$
E. None of the above

12. Computer Number Systems

In the octal representation for every year from 2001 to 2021 inclusive, how many total digits are greater than 4?

A. 25
B. 26
C. 27
D. 28
E. None of the above

13. Data Structures

What nodes (left to right) are at a depth of 2 in the min-heap created for the string:

TRICERATOPS

A. CERA
B. OERA
C. OIRR
D. TIRR
E. None of the above

14. Data Structures

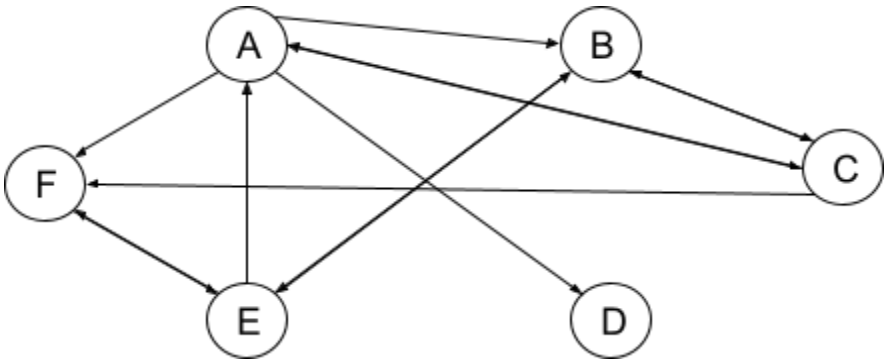
Accountants use two ways of accounting for calculating taxes on stock distributions: LIFO and FIFO. Suppose you bought 100 shares of a stock in each of the months March, April, and May for $10 per share, $15, and $5 respectively. If you sold 100 shares in July when the price was $20 per share, that means you now have 200 shares left. With FIFO accounting, you would calculate the profit based on the 100 shares you bought in March. With LIFO accounting, you would calculate the profit based on the 100 shares you bought in May.

Calculate the total profit each way and determine which method would be to your advantage as the taxpayer (i.e. less profit to pay taxes on) and how much would you save if the rate is 20% on all profits?

A. FIFO, saving $100
B. LIFO, saving $100
C. FIFO, saving $500
D. LIFO, saving $500
E. No difference

15. Graph Theory

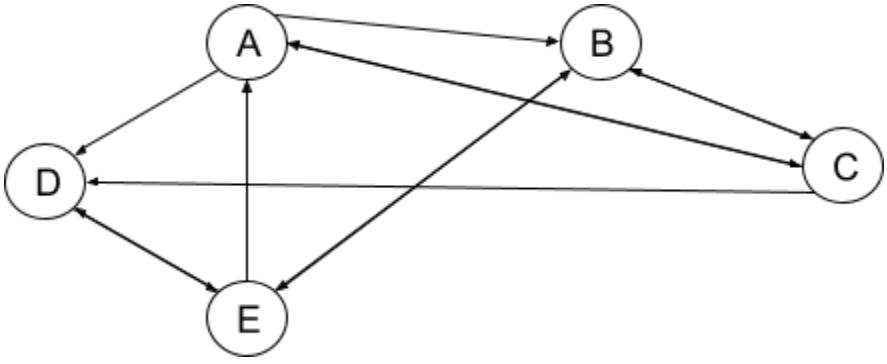
Given the directed graph below, how many unique pairs of vertices do not have a path of length 2 in either direction? (X,X) could be included.



A. 3
B. 4
C. 5
D. 6
E. None of the above

16. Graph Theory

How many cycles are there in the directed graph below?



A. 8
B. 9
C. 10
D. 11
E. None of the above

17. What Does This Program Do?

Pascal's Triangle is created as follows:

```
            1
          1   1
        1   2   1
      1   3   3   1
    1   4   6   4   1
```

Given that row X (starting at row 0) is stored in a one-dimensional array named **pt** of X+1 items, and the first (location 0) and last (location X) items are always assigned the value 1, which of the following loops will create the other values in row X+1 successfully?

A. for i = 1 to X
    pt(i) = pt(i) + pt(i-1)
  next i

B. for i = X to 1 step -1
    pt(i) = pt(i) + pt(i-1)
  next i

C. for i = X to 1 step -1
    pt(i+1) = pt(i) + pt(i-1)
  next i

D. for i = 1 to X
    pt(i-1) = pt(i-1) + pt(i)
  next i

E. None of the above

18. LISP

After the following LISP program is run, what is the value of the CONS expression?

```
(SETQ S '(H ((A C) K) (E (R R (A)))(N (K))))
(SETQ G (CADDADADDR S))
(SETQ Q '((P L) (A (T (F O) R)) M))
(SETQ H (CADADADR Q))
(CONS G H)
```

A. ((A) F O)
B. (A F O)
C. ((A) (F O))
D. (A (F O))
E. None of the above

19. FSA and Regular Expressions

A new school supply store just opened and is offering a 50% discount on any item that matches the following regular expression:

```
[^aeiou][aeiou][^a-j][p-t]?.(er|s)*
```

How many different items will be discounted from the following items?

crayons	markers	rulers	erasers
staples	staplers	notebooks	desks
computers	tablets	boards	compass

A. 7
B. 6
C. 5
D. 4
E. None of the above

20. Assembly Language

What is printed when the following program is run if the data is 8 and 3?

```
        READ    X
        READ    Y
        LOAD    X
        SUB     Y
        STORE   Z
        LOAD    =1
        STORE   C
        STORE   B
        STORE   A
T1      LOAD    C
        MULT    Z
        STORE   C
        LOAD    Z
        SUB     =1
        STORE   Z
        BG      T1
T2      LOAD    B
        MULT    Y
        STORE   B
        LOAD    Y
        SUB     =1
        STORE   Y
        BG      T2
T3      LOAD    A
        MULT    X
        STORE   A
        LOAD    X
        SUB     =1
        STORE   X
        BG      T3
        LOAD    A
        DIV     B
        DIV     C
        STORE   D
        PRINT   D
        END
```

A. 6,720
B. 336
C. 56
D. 24
E. None of the above